p-adic Representation Theory

— *EYNTKA* Series —

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Contents

0.0.1 Inspiration

Very heavily inspired by [2]

For a review of p-adic integers and p-adic fields (more generally of local fields), see [5, chapter 1.1].

I want here a rough outline of the Langlands program, in particular that $\mathrm{Gal}(\overline{K}/K)$ corresponds to complex representations of $\mathrm{GL}_n(\mathbb{Q}_p)$, and we want complex representations because this now would give us automorphic forms

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Smooth Representations

Recall that if $\mathrm{Gal}(K^{\mathrm{ab}}/K)$ is the maximal abelian extension of K, it is isomorphic to the profinite completion of the idele group $C_K = \mathbb{I}_K/K^*$. When $K = \mathbb{Q}$, this becomes the restricted product $\prod'_p \mathbb{Z}_p^*$ showing us that there are natural maps

$$\mathrm{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{Q}_p^* = \mathrm{GL}_1(\mathbb{Q}_p) \qquad \text{and} \qquad \mathrm{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{C}^* = \mathrm{GL}_1(\mathbb{C})$$

i.e. one-dimensional representations (over p-adic fields or $\mathbb{C}$). When $\mathrm{Gal}(L/K)$ is non-abelian, we get higher-dimensional. In this chapter we shall specifically focus on *complex representation*, as these shall are the object in the Langlands correspondence. In ref:HERE, we shall consider p-adic representations.

We start with looking at locally profinite groups in general $\mathbb{Q}_p$ and the multitude of natural objects formed by it are locally profinite. We then look into the right representation of locally profinite groups, in particular we shall require the right notion of smooth representation (as $\mathbb{Q}_p$ is far from being a smooth manifold) that parallels the process for the representation of Lie groups and has the same nice harmonic-analysis properties.

finish explaining here.

1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [4, chapter 8].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|-|_p$. This is a substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

Definition 1.1.1: P-adic Topology

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime p and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \cdots$$

Giving $\mathbb{Z}_p$ a p-adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|-|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p \mid |x-a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left|\frac{m}{n}\right|_p = \frac{|m|_p}{|n|_p}$$

Let us establish some basic properties of this field.

Lemma 1.1.2: P-adic Field Totally Disconnected

The field $\mathbb{Q}_p$ with the topology induced by the metric $|-|_p$ is totally disconnected. More generally any space X whose topology is induced by an ultra-metric is totally disconnected.

***Proof*:**
Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a d such that:

$$0 < |a-b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p \mid |x-a| < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p \mid |x-a| \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a-x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x-y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y-a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points a, x, y two of $d(a,x)$, $d(a,y)$ or $d(x,y)$ must be to the larger one. As $|a-x|_p \geq d$, $|y-a| \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

Lemma 1.1.3: P-adic Integers Open Subgroup

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be closed, hence connected, and hence $H = \mathbb{C}$. As is $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

***Proof*:**
First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p \mid |x - a|_p < 1\}$$

I claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.

Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

There shall be instances where we work with other similar groups constructed from p-adic fields, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L_\mathfrak{p}/\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$, $\mathbb{C}_p$, $\mathrm{GL}_n(\mathbb{Q}_p)$ or $\mathrm{GL}_n(\mathcal{O}_{L/\mathbb{Q}_p})$ (or matrix subgroups of this group), $\mathrm{GL}_n(\mathfrak{m}_\mathfrak{p})$ and so forth. Thus, it is worth going over a general theory that encapsulates the topology of these spaces. Let us first recall some basic properties of topological groups:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open
- If $H \leqslant G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- If K_1, K_2 are compact subset of G, so is K_1K_2
- subgroups in relation to open/closed:
  - if $H \leqslant G$ is open, it must be closed ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the compliment of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U, then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. We shall frequently be working with totally disconnected groups.
  - A closed subgroup $H \leqslant G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). However if $H \leqslant G$ is closed and finite index, it is open (the finite union of closed is closed). If G is first-countable, H is closed if it contains all its limit points (more generally, it it contains all the limits of nets).
  - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).
  - if H is a subgroup of G, so it $\overline{H}$ (the closure under limits/nets).
  - IF $G_0 \subseteq G$ is the connected component of the identity, then G_0 is closed, G_0 is the intersection of all normal subgroups of G, and G/G_0 is the largest totally disconnected quotient.
- symmetric neighbourhood:
  - For every open neighborhood of e, say for example U, there is a symmetric neighborhood (if $A \subseteq G$ such that $A = A^{-1}$, we'll say A is *symmetric*) V of e such that $V \subseteq U$
  - For every neighborhood U of e, there is a neighborhood V of e with $VV \subseteq U$

Lemma 1.1.4: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. H is a subgroup
2. H is the closure of e, i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

***Proof*:**
exercise

Henceforth, we shall always assume Hausdorffness unless specified otherwise.

Definition 1.1.5: Profinite Group

Take a diagram finite groups with the discrete topology. Then a *profinite group* is a topological group that is the colimit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \to \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.6: Profinite Groups

We have the following “trivial” examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the “usual” example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \cdots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005 \bmod 7^n$ for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the profinite completion of galois Galois groups, namely $\mathrm{Gal}_K(\overline{K})$. See [6] or a quick overview or [1] for more details.

More generally, profinite groups can be seen as the object that captures all the finite behaviour "up to infinity" (or philosophically, all the "potential infinity") of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group, see ref:HERE.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.7: Pro-finite Completion Injective

Let G be a topological group and $\widehat{G}$ it's profinite completion. Then $G \to \widehat{G}$ is injective if the intersection all finite-index open normal subgroups of G is trivial

***Proof*:**
exercise.

The completion comes equipped with a natural universal property:

Theorem 1.1.8: Universal Property of Profinite Completion

let $\varphi : G \to H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[dr, "\varphi"] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

***Proof*:**
exercise (hint: recall G is dense in $\widehat{G}$).

A striking result is that all profinite groups can be characterized by a simple topological property:

Theorem 1.1.9: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose G is totally disconnected and compact. We must show it comes from an inverse system of finite groups. will finish later.

Corollary 1.1.10: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all *open* normal subgroups ordered by contain need.
If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
exercise.

Definition 1.1.11: Locally Profinite Group

A topological group G is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G.

Notice how there is no mention of an inverse system. The reader should show that the definition imples G is totally disconnected, and hence if it is also compact then we recover a profinite group. Every discrete group is finite, closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite subgroup is locally compact. It is an exercise to show that any locally compact totally disconnected group is locally profinite. The canonical example for these notes of a locally profinite group is the non-Archimedean local field $(F, +)$ as proven in [5]. Similarly, $(F^\times, \times)$ is a profinite group by passing through e^x, or:

$$U_F^n = 1 + \mathfrak{p}^n$$

Being profinite is closed under products, hence F^n is profinite, and $(M_n(F), +)$ is locally profinite under addition. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ under multiplication is an open topological subgroup. It is locally profinite as $\mathrm{GL}_n(\mathcal{O}_F)$ is profinite (it is compact as it is homeomorphic to a closed subset of $\mathcal{O}_F^{n^2}$ which is a product of compact groups, and it is open as $\mathcal{O}_F = \{x \in F \mid |x|_\mathfrak{p} \leq 1\}$ is an open set. Note that this was all done in coordinates, more generally this applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$.

Let us now consider character's on locally profinite groups:

Theorem 1.1.12: Continuous Character

let $\varphi : G \to \mathbb{C}^\times$ be a group homomorphism where G is locally profinite. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups (such as $\mathbb{Q}_p$), then $\mathrm{im}\varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

***Proof*:**
If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker\varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing φ is continuous. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U, we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker\varphi$. As K is open, $\ker\varphi$ contains an open subgroup and hence by topological group theory $\ker\varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^\times$. If K' is a compact subgroup of G, $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

This condition has interesting algebraic consequences, for example take the map $\mathbb{Z} \to S^1$ by sending 1 to $e^{2\pi i\theta}$ where θ is irrational. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, this map extends to a map $\mathbb{Z}_p \to S^1$. It's kernel is $\{0\}$, i.e. it is not open, and so this map is not continuous with the profinite topology on $\mathbb{Z}_p$.

Corollary 1.1.13: Continuous Maps To General Linear Group

Let $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ be a group homomorphism where G is locally profinite. Then φ is continuous if and only if the kernel of φ is open.

If G is the union of compact open subgroup, the maps into $U(n)$ I'll prove this later.

***Proof*:**
Note that $\mathrm{GL}_n(\mathbb{C})$ is a lie group and that there exists an open neighbourhood of I_n that contains no non-trivial subgroups, the argument follows.

Definition 1.1.14: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \to \mathbb{C}^\times$$

We shall often write 1_G or just 1 for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character.*

summary of next section On page 12 in [2], he defines $\widehat{F}$ to be the set of all characters of F. As F is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ .Fixing a d, the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: "Useless" Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \to \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when G is not compact, we get a vector space V that is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators; this thread is further explored in [3].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness is different from smoothness for manifolds as we will be looking at maps $\mathbb{Q}_p \to \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in close will get the same linear, or "constant" if we change coordinates in our perspective. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \to \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$.

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so we'll assume [4] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.

2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support. The Fourier transform on $\mathbb{Q}_p$ is a good map on this space: the transform of a locally constant, compactly supported function is another locally constant, compactly supported function. See this link for more details.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \to \mathbb{C}^n$ a map. Then f is smooth if and only if for *any* subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous with $\mathbb{C}^n$ having the discrete topology.

***Proof*:**
exercise.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p-adic valuation $|-|_p : \mathbb{Q}_p^\times \to \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n\mathbb{Z}_p^\times$ which is open.

   Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

Often, a smooth function should be thought of as being well-defined modulo cosets of an open subgroup.

By identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, we get the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *complex representation* (or just *representation*) of G is a pair (π, V) where V is a complex vector space and $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $x \in V$, there is a compact open subgroup $K_x \subseteq G$ such that for all $k \in K_x$:

$$\pi(k) \cdot x = x$$

Equivalently, if V^K denotes the space of $\pi(K)$-fixed vector in V,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G.

A smooth representation is called *admissible* if V^K is finite dimensional for all compact open $K \subseteq G$.

Proposition 1.2.7: Smooth Representation Alternative

A representation $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G \mid \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

***Proof*:**
exercise.

If (π, V) is a smooth representation of G, then any G-stable subspace of V is another smooth representation, and similarly if $U \subseteq V$ is a G-subspace, the induced representation of G on V/U is smooth. A representation (π, V) is *irreducible* if if $V \neq 0$ and V has no nontrivial G-stable subspaces. A G-homomorphism between two representations (π_1, V_1), (π_2, V_2) is a map that commutes with the group action induce by π_i.

Example 1.2.8: Smooth Representations

1. If $(\mathbb{C}, \chi)$ is a one-dimensional representation, by theorem 1.1.12 it is smooth if and only if it is continuous. Hence in one dimensions these two concepts are the same. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and v is nonzero we have:

$$\chi(g) = 1$$

Hence, χ is smooth if and only if the kernel is open (and as G is locally profinite it contains a compact subgroup), which by theorem 1.1.12 is true if and only if χ is continuous.

By corollary 1.1.13, any finite dimensional continuous representation is smooth.

2. Let G be compact (thus profinite) and (π, V) an irreducible representation of G. Then V must be finite-dimensional. Indeed, if $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As G is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v \mid g \in G/K\}$$

spans V. We can then take the smaller $K' = \bigcup_{g \in G/K} gKg^{-1}$ which is an open normal subgroup of G, still of finite index, and the action is invariant on K'. Thus, V can be seen as an irreducible representation of the finite discrete group G/K', and hence is certainly finite dimensional.

3. Let $G = \mathbb{Q}_p^*$; let us look for maps $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ which as $\mathbb{Q}_p$ is the union of compact open sets by theorem 1.1.12 we know every maps into S^1. As $\mathbb{Q}_p^* \cong \mathbb{Z} \times \mathbb{Z}_p^*$ (since $x = p^n u$ for $n \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^*$; namely $n = \nu_p(x)$), and ψ is a group homomorphism, the map is determined by where it maps $(1, 0)$ and $\mathbb{Z}_p^*$. Let us look at the two parts seperatly:

$$\psi((c, u)) = \psi((c, 1)) \cdot \psi((0, u)) = \psi_1(c) \cdot \psi_2(u)$$

Thus, for any choice of $s \in \mathbb{C}$, let:

$$\psi_1(\nu_p(x)) = p^{-s\nu_p(x)}$$

Next, lets look at the units. Note that this includes $1 \in \mathbb{Z}_p^*$, and we know that the kernel must be open, and as $\varphi(1) = 1$, that means that an open neighbourhood around $1 \in \mathbb{Z}_p^*$ maps to $1 \in \mathbb{C}^*$. The open neighborhoods around 1 are:

$$U_k = 1 + p^k \mathbb{Z}_p = \left\{ u \in \mathbb{Z}_p^* \;\middle|\; u \equiv 1 \bmod (p^k) \right\}$$

Thus, for some $k \in \mathbb{N}$ $\psi_2(U_k) = 1$ By the isomorphism theorem the cosets will be:

$$\mathbb{Z}_p * /U_K \cong (\mathbb{Z}/p^k\mathbb{Z})^*$$

But now ψ_2 reduces to $(\mathbb{Z}/p^k\mathbb{Z})^* \to \mathbb{C}^*$ which is just the familiar roots of unity maps. Note how this is "ramified". For this reason, ψ_1 is often called the *unramified component* while ψ_2 is called the *ramified component*.

Combining these together, if we let $\psi_1(x) = |x|_p^s$ and $\psi_2(x) = \psi(x)$:

$$\chi(x) = |x|_p^s \psi(x)$$

These classify all continuous/smooth characters of $\mathbb{Q}_p^*$.

4. Let us next consider the next dimension up: $G = \mathrm{GL}_n(\mathbb{Q}_p)$ an $V = \mathbb{C}$, that is smooth characters on $\mathrm{GL}_n(\mathbb{Q}_p)$. Then As $\mathbb{C}^*$ is commutative, $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$ must be trivial on the commutator. It is an exercise in linear algebra to show that $[G, G] = \mathrm{SL}_n(\mathbb{Q}_p)$[a]. As the identity has determinant 1, $I \in \mathrm{SL}_n(\mathbb{Q}_p)$ and as χ is a group homomorphism it must map to 1. Thus $\chi(\mathrm{SL}_n(\mathbb{Q}_p)) = 1$.

Next, for any $A, B \in \mathrm{GL}_n(\mathbb{Q}_p)$ such that $\det(A) = \det(B)$ so that $AB^{-1} \in \mathrm{SL}_n(\mathbb{Q}_p)$, we must have $\chi(A) = \chi(B)$ as χ is a homomorphism. But then, χ is only dependent on the determinant of a matrix.

From this, to can construct a character $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$, we choose a 1-dimensional character $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ and compose it with the determinant map $A \mapsto \det(A)$ so that:

$$\chi(A) = \psi(\det(A))$$

It can be shown that all characters are of this form.

5. Let us consider $G = \mathbb{Q}_p$ and $V = \mathbb{C}$ so that we are looking at characters $\chi : \mathbb{Q}_p \to \mathbb{C}^*$. These are all formed by the *standard character*. For any $x \in \mathbb{Q}_p$ the p-adic representation is of the form
$$x = \sum_{n}^{\infty} a_i p^i \qquad n \in \mathbb{Z},\ a_i \in \{0, 1, ..., p-1\}$$
The p-adic fractional part is the negative indexed summands; let $\{x\}_p$ represent this. Define:
$$\psi_p : \mathbb{Q}_p \to \mathbb{C}^* \qquad \psi_p(x) = e^{2\pi i \{x\}_p}$$
Notice the kernel of this map is $\mathbb{Z}_p$, and hence it is continuous/smooth.

6. We can form characters on $\mathfrak{m}_p$ by restricting characters on $\mathbb{Q}_p$. Show that $\hat{\mathfrak{m}}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$.

7. For a continuous non-smooth example, let $G = \mathbb{Z}_p$ and $V = C(\mathbb{Z}_p, \mathbb{C})$ be the $\mathbb{C}$-vector space of continuous characters with the uniform-convergence topology. Then take the left-regular representation (ρ, V):
$$(\rho(g) \cdot f)(x) = f(x - g)$$
The reader can show this is continuous. Let us look at the stabilizer of the vector $g \mapsto |g|_p$. Note this map is certainly continuous as the norm-map is continuous, and thus $| - |_p \in C(\mathbb{Z}_p, \mathbb{C})$. Let us consider the stabilizer of $| - |_p$ that is all, $g \in \mathbb{Z}_p$ such that :
$$g \cdot | \bullet |_p = | \bullet - g|_p = | \bullet |_p$$
Let $g \in \mathrm{Stab}_{\mathbb{Z}_p}(| - |_p)$ so that:
$$|g - x|_p = |x|_p \forall x \in \mathbb{Z}_p$$
If $x \neq g$, then it easy to see that equality *fails*, hence the only stabilizer is 0. But then $\mathrm{Stab}_{\mathbb{Z}_p}(| - |_p) = \{0\}$ which is *not* open.

8. (principal series representation) Let:
$$B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}_p)) \} \right\}$$
Let us define a character on B by choosing two characters $\chi_1, \chi_2 : \mathbb{Q}_p^* \to \mathbb{C}^*$ given by:
$$\varphi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = \chi_1(a)\chi_2(c)$$
Now, define the complex vector space V to be smooth complex function $f : \mathrm{GL}_2(\mathbb{Q}_p) \to \mathbb{C}$ such that:
$$f(bg) = \psi(b) f(g) \qquad b \in B,\ g \in \mathrm{GL}_2(\mathbb{Q}_p)$$
Then define the action to be:
$$(\rho(h) \cdot f)(g) = f(gh)$$
Then this is an infinite dimensional smooth representation; for a choice of χ_1, χ_2 denote it $I(\chi_1, \chi_2)$. It is almost always irreducible, but there shall be there are non-reducible representations given the right choices of χ_1, χ_2, see example 1.2.12.

By Schur's Lemma, the abelian groups have no higher dimensional irreducible representations. For higher finite dimensional smooth representations of the non-abelian groups, ref:HERE shall show that they are *all* a direct sum of 1 dimensional representations, even though these groups are not abelian! This shows that to capture some of their more non-abelian structure of these groups, it will be important to look at (well-behaved) infinite dimensional representations, hence the need to add the smooth condition.

[a]this is true as long as the field F is not the field with two elements

As there are many hierarchies of types of representations in representation theory (semi-stable, crystalline, etc), we shall purposefully specify the category that has been described thus far:

Definition 1.2.9: Smooth Representation Category

Let G be a locally profinite group. Then $\mathrm{Rep}_{\mathrm{smooth},K}(G)$ is the smooth representation category whose objects are smooth representations (V,π) and morphisms are G-equivariant maps. The representations shall always be smooth, and hence we write $\mathrm{Rep}_K(G)$ instead of $\mathrm{Rep}_{\mathrm{smooth},K}(G)$. The field K is dropped if it is clear from context; by default in this book it shall be $\mathbb{C}$.

By a similar argument as the usual category for representations of finite groups, $\mathrm{Rep}(G)$ is an abelian category. When working with finite groups, we saw that representations can be thought of as $\mathbb{C}[G]$-modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$-module V than smooth representations; we shall need to limit to *Hecke algebras* as shall be discussed in section 1.4.

Let us next upgrade the standard decomposition result from finite group representation theory:

Proposition 1.2.10: Semisimple Equivalence

Let G be a locally profinite group, and let (π, V) be a smooth representation of G. The following conditions are equivalent:

1. V is the sum of its irreducible G-subspaces;
2. V is the direct sum of a family of irreducible G-subspaces;
3. any G-subspace of V has a G-complement in V.

***Proof*:**
(1) $\Rightarrow$ (2). Take the set of irreducible G-subspaces U_i of V

$$\{U_i : i \in I\}$$

such that $V = \sum_{i\in I} U_i$. Consider the set $\mathcal{I}$ of subsets J of I such that the sum $\sum_{i\in J} U_i$ is direct. The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a\in A} J_a$. If the sum $\sum_{j\in J} U_j$ is not direct, there is a finite subset S of J for which $\sum_{j\in S} U_j$ is not direct. Since S must be contained in some J_a, this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element

J_0 of $\mathcal{I}$. For this set, we have

$$V = \bigoplus_{a \in J_0} U_i,$$

as required for (2).

(2) $\Rightarrow$ (3). In (3), let W be a G-subspace of V. By (2), we can assume that $V = \bigoplus_{i \in I} U_i$, for a family (U_i) of irreducible G-subspaces of V. We consider the set $\mathcal{J}$ of subsets J of I for which $W \cap \sum_{i \in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If J is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j \in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1) Let V_0 be the sum of all irreducible G-subspaces of V and write $V = V_0 \oplus W$, for some G-subspace W of V. Assume for a contradiction that $W \neq 0$. By its definition, the space W has no irreducible G-subspace. However, there is a non-zero G-subspace W_1 of W which is finitely generated over G. By Zorn's Lemma, W_1 has a maximal G-subspace W_0, and then W_1/W_0 is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some G-subspace U of V, and hence a G-projection $V \to U$. The image of W_1 in U is an irreducible G-subspace of U, hence an irreducible subspace of V which is not contained in V_0. This is nonsense, so $V = V_0$ and (3) $\Rightarrow$ (1).

Definition 1.2.11: Semisimple Representation

A representation (π, V) is *G-semisimple* if it satisfies one of the equivalent conditions of proposition 1.2.10.

Unlike finite groups, locally profinite groups will have (many) non-semisimple complex representations:

Example 1.2.12: Reducible non-semisimple representation

The details will be left for the reader. Let $\chi_1(x) = |x|_p$ and $\chi_2(x) = |x|_p^{-1}$ and take the principal series representation $I(\chi_1, \chi_2)$. This representation has a unique 1-dimensional subrepresentation which is a character of $\mathrm{GL}_2(\mathbb{Q}_p)$; denote it χ. The quotient of $I(\chi_1, \chi_2)$ by this representation is called the *Steinberg representation* and is denoted St. This fit into a short exact sequence:

$$0 \to \chi \to I(\chi_1, \chi_2) \to \mathrm{St} \to 0$$

However, this sequence *does not split*. Thus, it is *reducible* but *indecomposable*, hence *not semisimple*.

The compact opens subgroups shall fair better:

Lemma 1.2.13: Compact Open Semistable Representation

Let G be a locally profinite group, and let K be a compact open subgroup of G. Let (π, V) be a smooth representation of G. The space V is the sum of its irreducible K-subspaces:

$$V = \sum_{\substack{W \subseteq V \\ \text{irred. } K\text{-subspace}}} W$$

We say that V is *K-semisimple*.

***Proof*:**
Let $v \in V$. As in example 1.2.8, v is fixed by an open normal subgroup K' of K, and it generates a finite-dimensional K-space W on which K' acts trivially. Thus W is effectively a finite-dimensional representation of the finite group K/K' and so is the sum of its irreducible K-subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Thus, the following is a meaningful definition:

Definition 1.2.14: Isotypic Component

Let G be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible representations. Let $\rho \in \widehat{K}$ and (π, V) be a smooth representation of G. Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the ρ-*isotypic component* of V.

By definition, V^K is the isotypic subspace for the class of trivial representation 1 of K.

Proposition 1.2.15: K-isotypic Decomposition

Let (π, V) be a smooth representation of G and let K be a compact open subgroup of G.

1. The space V is the direct sum of its K-isotypic components:
$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho.$$

2. Let (σ, W) be a smooth representation of G. For any G-homomorphism $f : V \to W$ and $\rho \in \widehat{K}$, we have
$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho).$$

***Proof*:**
By proposition 1.2.10:
$$V = \bigoplus_{i \in I} U_i,$$
for a family of irreducible K-subspaces U_i of V. We let $U(\rho)$ be the sum of those U_i of class ρ. We then have
$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho).$$

If W is an irreducible K-subspace of V of class ρ, then $W \subset U(\rho)$; otherwise, there would be a non-zero K-homomorphism $W \to U_i$, for some U_i of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of V^ρ is a sum of irreducible K-subspaces of W, all of class ρ and therefore contained in W^ρ. Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the W^τ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

We frequently use part (2) of the Proposition in the following context:

Corollary 1.2.16: Exactness reduced to compact subgroup

Let $a : U \to V$, $b : V \to W$ be G-homomorphisms between smooth representations U, V, W of G. The sequence

$$U \xrightarrow{a} V \xrightarrow{b} W$$

is exact if and only if

$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$

is exact, for every compact open subgroup K of G.

***Proof*:**
exercise.

If H is a subgroup of G, we define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}. \tag{2.3.1}$$

In particular, $V(H)$ is an H-subspace of V.

Corollary 1.2.17: Splitting Using Compact Subgroups

Let G be a locally profinite group, and let (π, V) be a smooth representation of G. Let K be a compact open subgroup of G. Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique K-complement of V^K in V.

***Proof*:**
The sum $W = \bigoplus V^\rho$, with ρ not trivial, is a K-complement of V^K in V. There is therefore a K-surjection $V \to V^K$ with kernel W. Clearly, $V(K)$ is contained in the kernel of any K-homomorphism $V \to V^K$; we conclude that W contains $V(K)$. On the other hand, if U is an irreducible K-space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

The next important concept is that of induced representation and how we can construct representations using them. Recall that in the finite case a subgroup $H \subseteq G$ which has a representation (W, σ) will have a natural induced representation $(\mathbb{C}[G]_{\mathbb{C}[H]}W, \mathrm{Ind}_H^G \sigma)$ where the space $\mathbb{C}[G]_{\mathbb{C}[H]}W$ can be represented as H-equivariant functions and Ind_H^G acts on the set of these functions. The following generalizes this:

Definition 1.2.18: Induced Representation

Let G be a locally profinite group, and let $H \subseteq G$ be a closed subgroup of G so that H is also locally profinite. Let (σ, W) be a smooth representation of H.

Define the space X of functions $f : G \to W$ which satisfy:

1. equivariant: $f(hg) = \sigma(h)f(g)$, $h \in H$, $g \in G$
2. smoothness: there is a compact open subgroup K of G (depending on f) such that $f(gx) = f(g)$, for $g \in G$, $x \in K$ (so f is smooth i.e. continuous with W having the discrete topology)

Then define a homomorphism $\Sigma : G \to \mathrm{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \longmapsto f(xg), \quad g, x \in G.$$

The pair (Σ, X) provides a smooth representation of G and is said to be *smoothly induced by* σ. It is usually denoted

$$(\Sigma, X) = \mathrm{Ind}_H^G \sigma.$$

Proposition 1.2.19: Induced Representation is A Functor

The map $\sigma \mapsto \mathrm{Ind}_H^G \sigma$ gives a functor $\mathrm{Rep}(H) \to \mathrm{Rep}(G)$.

***Proof*:**
exercise

There is a canonical H-homomorphism

$$\alpha_\sigma : \mathrm{Ind}_H^G \sigma \longrightarrow W, \quad f \longmapsto f(1).$$

The pair $(\mathrm{Ind}_H^G, \alpha)$ has the following fundamental property:

Theorem 1.2.20: Frobenius Reciprocity

Let H be a closed subgroup of a locally profinite group G. For a smooth representation (σ, W) of H and a smooth representation (π, V) of G, the canonical map

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \sigma) \longrightarrow \mathrm{Hom}_H(\pi, \sigma)$$
$$\varphi \mapsto \alpha_\sigma \circ \varphi,$$

is an isomorphism that is functorial in both variables π, σ.

***Proof*:**
Let $f : V \to W$ be an H-homomorphism. We define a G-homomorphism $f_* : V \to \mathrm{Ind}\sigma$ by letting $f_*(v)$ be the function $g \mapsto f(\pi(g)v)$. The map $f \mapsto f_*$ is then the inverse of the above map.

The following shows how this is a very useful functor:

Proposition 1.2.21: Induced Representation is Additive and Exact

The functor $\mathrm{Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

need to re-phrase this!

***Proof*:**
For a smooth representation (σ, W) of H, temporarily let $I(\sigma)$ denote the space of functions $G \to W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus I is a functor to the category of abstract representations of G; it is clearly additive and exact, while $\mathrm{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus Ind_H^G is surely additive, and 2.3 Exercise (3) shows it to be left-exact.

To prove it is right-exact, let (σ, W), (τ, U) be smooth representations of H and let $f : W \to U$ be an H-surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup K of G which fixes φ. The support of φ is a union of cosets HgK, and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By 2.3 Corollary 1 (applied to the group H and the trivial representation of its compact open subgroup $H \cap gKg^{-1}$), there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \to W$ to have the same support as φ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H\backslash \mathrm{supp}\varphi/K$. The function Φ is fixed by K, and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is φ, as required.

Example 1.2.22: Induced Representation

1. here
2. here

As the compact open subgroups are the key subgroups to consider for this book, there is the following useful variation of induced representation.

Definition 1.2.23: Compact Induction

Given (σ, W) for a representation of closed $H \subseteq G$ and similarly defined X as in definition 1.2.18, define:

$$X_c = \{f \in X \mid \text{compactly supported modulo } H\}$$

that is the image of $\mathrm{supp}(f)$ in $H \setminus G$ is compact (equivalently, $\mathrm{supp}(f) \subseteq HC$ for some compact set $C \subseteq G$). The induced functor is denoted $\text{c-Ind}_H^G\sigma$.

Proposition 1.2.24: Compact Induction is Well-defined

The space X_c along with $\text{c-Ind}_H^G\sigma$ is a smooth represetnation

***Proof*:**
exercise.

Proposition 1.2.25: Compact Induction Repreis Additive and Exact

The functor $\text{c-Ind}_H^G : \text{Rep}(H) \to \text{Rep}(G)$ is additive and exact.

***Proof*:**
exercise.

Example 1.2.26: Compact Induction

1. here
2. here

There is a natural G-embedding:

$$\text{c-Ind}_H^G(\sigma) \to \text{Ind}_H^G(\sigma) \qquad \text{(i.e. a natural transformation } \text{c-Ind}_H^G \Rightarrow \text{Ind}_H^G)$$

This map is an isomorphism if and only if $H \setminus G$ is compact. [2] says that there are examples of such a map being an isomorphism even when $H \setminus G$ is not compact with the right choice of G, H, σ.

The map is mostly interesting when $H \subseteq G$ is open in which case there is a natural H-homomorphism:

$$\alpha_\sigma^c : W \to \text{c-Ind}(\sigma) \qquad w \mapsto f_w$$

go over the Frobenius for compact; skipping for now because I have too little intuition

1.2.27 Hypothesis From Now On For any compact open $K \subseteq G$, G/K is countable.

If G/K is countable, so is G/K' for any other compact open; for $K \cap K'$ is compact open and a finite index of K thus

$$\frac{G}{K \cap K'} \to G/K$$

has finite fibers, thus $G/K \cap K'$ and G/K' are both countable. All groups that will be considered in this book are have this property, but a few further results bellow don't require them. The main point is so that:

Lemma 1.2.28: Irreducible Smooth Representation At Most Countable

Let (π, V) be a smooth irreducible representation of a locally profinite group G. Then $\dim_{\mathbb{C}}(V)$ is countable

***Proof*:**
Let $0 \neq v \in V$ and choose $K \subseteq G$ such that $v \in V^K$. As V is irreducible,

$$\{\pi(g) \cdot v \mid g \in G/K\}$$

spans V, and this set is countable completing the proof.

This also allows for the following:

Theorem 1.2.29: Schur's Lemma For Smooth Representations

Let (π, V) be a smooth irreducible representation of a locally profinite group G. Then

$$\mathrm{End}_{\mathbb{C}}(V) = \mathbb{C}$$

***Proof*:**
Let $0 \neq \varphi \in \mathrm{End}_{\mathbb{C}}(V)$. As the image and kernel of φ must be G-subspaces, by irreducibility it must be bijective (which is enough in this category for invertible). Thus, $\mathrm{End}_{\mathbb{C}}(V)$ is at least a division $\mathbb{C}$-algebra.

Next, $\mathrm{End}_{\mathbb{C}}(V)$ must have countable dimension for if $0 \neq v \in V$, the G-translate of v spans V, and so $\varphi \in \mathrm{End}_{\mathbb{C}}(V)$ is uniquely determined by the value of $\varphi(v)$. Now, if $\varphi \in \mathrm{End}_{\mathbb{C}}(V)$ where $\varphi \tilde{\in} \mathbb{C}$, φ would be transcendental over $\mathbb{C}$ and hence the subset

$$\left\{ (\varphi - a)^{-1} \;\middle|\; a \in \mathbb{C} \right\}$$

would be linearly independent, and hence $\mathrm{End}_{\mathbb{C}}(V)$ would have uncountable $\mathbb{C}$-dimension, a contradiction. Thus, $\mathrm{End}_{\mathbb{C}}(V) \cong \mathbb{C}$, as we sought to show.

Note that this is an if and only if when working with finite groups. If G is compact, this becomes an if and only if. This is false in general ([2] cites section 9.10).

Corollary 1.2.30: Central Character

Let (π, V) be an irreducible smooth representation of G and $Z = Z(G)$ is the center of G. Then Z acts on V via the character

$$\omega_\pi : Z \to \mathbb{C}^* \qquad \text{i.e.} \qquad \pi(z) \cdot v = \omega_\pi(z) \cdot v$$

for all $v \in V$ and $z \in Z$

***Proof*:**
As $z \in G$, the left-multiplication map is an endomorphism, hence $\pi(z) \in \mathrm{End}_G(V) \cong G$. Thus $\pi(z)$ is some scalar multiple of id_V, hence $\pi(z) = \omega_\pi(z)$. To shows it's a character, note that if K is a compact open subgroup of G such that $V^K \neq 0$, Then ω_π is trivial om the compact open subgroup $K \cap Z$, showing it is indeed a smooth representation, and hence ω_π is a character of Z.

Definition 1.2.31: Central Character

The function $Z \to \mathbb{C}^*$ defined in corollary 1.2.30 is called a *central character*.

Definition 1.2.32: Smooth Representation Admitting Central Character

Let (π, V) be a smooth representation of G and χ a central character of $Z(G)$. Then we say that (π, V) *admits* χ *as a central character* if:

$$\pi(z) \cdot v = \chi(z) \cdot v \qquad \forall v \in V,\ z \in V$$

Proposition 1.2.33: Irreducible Smooth Representation of Abelian Groups

Let G be an abelian locally profinite group. Then an smooth irreducible representation is one-dimensional

***Proof*:**
By corollary 1.2.30, any representation π will act like the character at every element $g \in G$ since $Z(G) = G$. Now, if W is any one-dimensional subspace, as π acts by scaling, W is a G-subspace. Hence, any irreducible subspace must be one-dimensional, completing the proof.

Proposition 1.2.34: KZ Space

Let (π, V) be a smooth representation of G, admitting χ as a central character. Let K be an open subgroup of G such that KZ/Z is compact. Then:

1. Let $v \in V$. The KZ-space spanned by v is of finite dimension, and is a sum of irreducible KZ-spaces.

2. As representation of KZ, the space V is semisimple.

***Proof*:**
The vector v is fixed by a compact open subgroup K_0 of K. The set KZ/K_0Z is finite, so the space W spanned by $\pi(KZ)v$ has finite dimension. Then W is K_0Z-semisimple. Since v was chosen arbitrarily, we get 2 as we sought to show.

The final elementary concept to adders is the dual space and how a representation (π, V) relates to it:

Definition 1.2.35: Smooth Dual (Contragradient)

here

this lemma is given next, may move it somewhere better. It is useful for the following discussion about smooth duals.

Lemma 1.2.36: Semisimple With Finite-index and Induction

Let G be a locally profinite group, and let H be an open subgroup of G of finite index.

1. If (π, V) is a smooth representation of G, then V is G-semisimple if and only if it is H-semisimple.
2. Let (σ, W) be a semisimple smooth representation of H. The induced representation $\mathrm{Ind}_H^G \sigma$ is G-semisimple.

***Proof*:**
Suppose that V is H-semisimple, and let U be a G-subspace of V. By hypothesis, there is an H-subspace W of V such that $V = U \oplus W$. Let $f : V \to U$ be the H-projection with kernel W. Consider the map

$$f^G : v \mapsto (G : H)^{-1} \sum_{g \in G/H} \pi(g) f(\pi(g)^{-1} v), \quad v \in V.$$

The definition is independent of the choice of coset representatives and it follows that f^G is a G-projection $V \to U$. We then have $V = U \oplus \mathrm{Ker}\, f^G$ and $\mathrm{Ker}\, f^G$ is a G-subspace of V. Thus V is G-semisimple (cf. 2.2 Proposition).

Conversely, suppose that V is G-semisimple. Thus G is a direct sum of irreducible G-subspaces (2.2), and it is enough to treat the case where V is irreducible over G. As representation of H, the space V is finitely generated and so admits an irreducible H-quotient U. Suppose for the moment that H is a normal subgroup of G. By Frobenius Reciprocity (2.4.2), the H-map $V \to U$ gives a non-trivial, hence injective, G-map $V \to \mathrm{Ind}_H^G U$. As representation of H, the induced representation $\mathrm{Ind}_H^G U = \oplus_{g \in G/H} gU$ is a direct sum of G-conjugates of U (cf. 2.5 Lemma). These are all irreducible over H, so $\mathrm{Ind}\, U$ is H-semisimple. Proposition 2.2 then implies that $V \subset \mathrm{Ind}\, U$ is H-semisimple.

In general, we set $H_0 = \bigcap_{g \in G/H} gHg^{-1}$. This is an open normal subgroup of G of finite index. We have just shown that the G-space V is H_0-semisimple; the first part of the proof shows it is H-semisimple.

This completes the proof of (1), and (2) follows readily from the same arguments.

1.3 Measures and Duality

The space $\mathbb{C}[G]$ is $\mathbb{C}$-algebra isomorphic to $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ where $*$ is convolution. This requires summing over each $g \in G$. In this book, G is a.e. infinite, and so we replace summation with an appropriate measure on G.

Let G be a locally profinite group, $C_c^\infty(G)$ the space of functions $f : G \to \mathbb{C}$ which are locally constant and of compact support (i.e. smooth and of compact support). Let $f \in C_c^\infty(G)$. Then by these properties there exists open subgroup $K_1, K-2 \subseteq G$ such that

$$f(k_1 g) = f(g) = f(g k_2) \qquad \forall g \in G,\ k_i \in K_i$$

By letting $K = K_1 \cap K_2$, f must be a finite linear combination of characteristic functions of double cosets KgK. Now, $G \curvearrowright C_c^\infty(G)$ by left translation and right translation:

$$\begin{cases} \lambda : g \mapsto (\lambda_g f : x \mapsto f(g^{-1}x)) \\ \rho : g \mapsto (\rho_g f : x \mapsto f(xg) \end{cases} \qquad x, g \in G,\ f \in C_c^\infty(G)$$

The reader should check that $(C_c^\infty(G), \lambda)$ and $(C_c^\infty(G), \rho)$ are both smooth G-representation.

Definition 1.3.1: Haar Integral

A *right Haar integral* on G is a non-zero linear functional:

$$I : C_c^\infty(G) \to \mathbb{C}$$

such that

1. $I(\rho_g f) = I(f)$ for all $g \in G,\ f \in C_c^\infty(G)$
2. For all $f \in C_c^\infty(G)$ where $f \geq 0$, $I(f) \geq 0$.

And similarly for the left Haar integral.

Proposition 1.3.2: Existence of Haar Integral

There exists right Haar integral $I : C_c^\infty(G) \to \mathbb{C}$. A linear functional $I' : C_c^\infty(G) \to \mathbb{C}$ is a right Haar integral if and only if $I' = cI$ for some $c > 0$.

The same then applies to the left Haar integral.

some parts of this proof reference results I did not write down yet; I'll put the label from the textbook for now.

***Proof*:**
Let K be a compact open subgroup of G Denote by ${}^K C_c^\infty(G)$ the space of functions in $C_c^\infty(G)$ fixed by $\lambda(K)$. Then viewing ${}^K C_c^\infty(G)$ as G-space via right translation, it is then identical to the representation of G compactly induced from the trivial representation 1_K of K:

$${}^K C_c^\infty(G) = c\text{-Ind}_K^G 1_K.$$

Now, viewing $\mathbb{C}$ as the trivial G-space, we have

$$\dim_{\mathbb{C}} \operatorname{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = 1.$$

and there exists a non-zero element $I_K \in \operatorname{Hom}_G({}^K C_c^\infty(G), \mathbb{C})$ such that $I_K(f) \geq 0$ whenever $f \geq 0$. If h_K is the characteristic function of K, then $I_K(h_K) > 0$.

Indeed, the first assertion is given by 2.5 Proposition. For $g \in G$, let f_g denote the characteristic function of Kg. The set of functions f_g, $g \in K\backslash G$, then forms a $\mathbb{C}$-basis of the space ${}^K C_c^\infty(G)$ (2.5 Lemma). The functional $I_K : f_g \mapsto 1$ has the required properties, noting that $h_K = f_1$.

Now, choose a descending sequence $\{K_n\}_{n\geq 1}$ of compact open subgroups K_n of G such that

$\bigcap_n K_n = 1$. Then:

$$C_c^\infty(G) = \bigcup_{n\geq 1} {}^{K_n}C_c^\infty(G).$$

For each $n \geq 1$, there is a unique right G-invariant functional I_n on ${}^{K_n}C_c^\infty(G)$ which maps the characteristic function of K_n to $(K_1 : K_n)^{-1}$. We have $I_{n+1}|_{{}^{K_n}C_c^\infty(G)} = I_n$, and so the family $\{I_n\}$ gives a functional on $C_c^\infty(G)$ of the required kind. The uniqueness statement is immediate, completing the proof.

Corollary 1.3.3: Left-Right Haar Integral

For $f \in C_c^\infty(G)$, define $\check{f} \in C_c^\infty(G)$ by $\check{f}(g) = f(g^{-1})$, $g \in G$. The functional

$$I' : C_c^\infty(G) \longrightarrow \mathbb{C},$$

$$I'(f) = I(\check{f}),$$

is a left Haar integral on G. Moreover, any left Haar integral on G is of the form cI', for some $c > 0$.

***Proof*:**
immediate.

From this, we can define a measure called the *Haar measure*. Let I be a left Haar integral on G and $\emptyset \neq S \subseteq G$ a compact open subset of a locally profinite group.

$$\mu_G(S) = I(\Gamma_S)$$

where Γ_S is the characteristic function of S. Then $\mu_G(S) > 0$ and μ_G is left-translation invariant:

$$\mu_G(gS) = \mu_G(S) \qquad \forall g \in G$$

The reader should check that this is indeed a measure (see [4] for a review of measure theory)

Definition 1.3.4: Haar Measure

The measure μ_G defined above is called the *Haar measure*.

The function I is often written as:

$$I(f) = \int_G f(g)d\mu_G(g) \qquad f \in C_c^\infty(G)$$

with the usual abbreviation of:

$$\int_S f(x)d\mu_G(x) := \int_G \Gamma_S(x)f(x)d\mu_G(x)$$

Definition 1.3.5: Unimodular Group

Let G be a locally profinite group (more generally a topological group with a Haar measure defined on it). Then G is *unimodular* if and only if the left Haar integral and is a right Haar integral.

Let us extend the domain of Haar integration (similarly to what is done to the Lebesgue measure). lots here; and relation between the Haar integral and character is outlined in detail.

An important result that is being build-up to:

Theorem 1.3.6: Duality Theorem

Let $\dot{\mu}$ be a positive semi-invariant measure on $H\backslash G$. Let (σ, W) be a smooth representation of H. Then there is a natural isomorphism:

$$(c\text{-Ind}_H^G \sigma)^\vee \cong \text{Ind}_H^G \delta_{H\backslash G} \otimes \check{\sigma},$$

depending only on the choice of $\dot{\mu}$.

***Proof*:**
will come back to.

1.4 Hecke Algebra

Linear representations of finite groups are in bijective correspondence to the modules over the group algebra $\mathbb{C}[G]$. A similar conditions holds for smooth representations of locally profinite groups given the suitable generalization of $\mathbb{C}[G]$ to Hecke algebras.

Though not necessary, to avoid technicalities that don't add insight, throughout this entire section the locally profinite group G will be assumed to be unimodular.

Fix a Haar measure μ on G. Then for any $f_1, f_2 \in C_c^\infty(G)$, define the convolution:

$$f_1 * f_2(g) = \int_G f_1(x) f_2(x^{-1}g) d\mu(x)$$

This gives a function $(x,) \mapsto f_1(x)f_2(x^{-1}g) \in C_c^\infty(G \times G)$ and so by ref:HERE $f_1 * f_2 \in C_c^\infty(G)$. This operation is associative: Here's the equation in an align* environment:

$$\begin{aligned}
f_1 * (f_2 * f_3)(g) &= \int\int f_1(x) f_2(y) f_3(y^{-1}x^{-1}g) d\mu(y) d\mu(x) \\
&= \int\int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(y) d\mu(x) \\
&= \int\int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(x) d\mu(y) \\
&= (f_1 * f_2) * f_3(g)
\end{aligned}$$

Definition 1.4.1: Hecke Algebra

Let G be a unimodular locally profinite group. Then:

$$\mathcal{H}(G) = (C_c^\infty(G), *)$$

is is an associative $\mathbb{C}$-algebra called the *Hecke algebra* of G.

This is an example of a *non-unital* algebra if G is not finite. If G is commutative, so is $\mathcal{H}(G)$. Note that if μ, ν are two choices of Haar measures giving rise to $\mathcal{H}_\mu(G)$ and $\mathcal{H}_\nu(G)$ on $C_c^\infty(G)$, then there exists a constant $c > 0$ such that $\nu = c\mu$ and a map $f \mapsto c^{-1} f$, and hence

$$\mathcal{H}_\mu(G) \cong \mathcal{H}_\nu(G)$$

The actual correspondence is proven here, going to write out the details later.

Goal: An $\mathcal{H}(G)$-module is smooth if $\mathcal{H}(G) * M = M$. If M is a smooth $\mathcal{H}(G)$-module, then there exists a homomorphism $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(M)$ such that (π, M) is a smooth representation. A vice-versa.

1.5 Important Special Case: Representation of $\mathrm{GL}_2(\mathbb{F}_p)$

When analyzing $\mathrm{GL}_2(\mathbb{Q}_p)$ and related groups, we shall draw inspiration of this special case. We shall first single out the following subgroups:

Definition 1.5.1: Matrix Subgroups

Let F be (an arbitrary) field. Then define the groups:

1. (Bruhat subgroup) $B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
2. (unipotent subgroup) $N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
3. (torus subgroup) $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
4. (center) $Z = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong F^*$

Lemma 1.5.2: Bruhat Decomposition

Let $G = \mathrm{GL}_2(F)$. Then its *Bruhat decomposition* is:

$$G = B \cup BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence, $\{1, w\}$ is a set of representative of the coset space $B \backslash G / B$.

Furthermore, $B = NT = TN$, hence $B \cong N \rtimes T$

***Proof*:**
Let $M \in GL_2(F)$ so that $\det(M) = ad - bc \neq 0$ if $c = 0$, $ad \neq 0$, so $a, d \neq 0$. Then $M \in B$.

If $c \neq 0$, choose two arbitrary $B_1, B_2 \in B$, calculate $B_1 w B_2$, use the fact that $\det(B_1), \det(B_2) \neq 0$ to get an equation for each entry of M. Note that the bottom-left entry *must* be nonzero and hence $B \cap BwB = \emptyset$, proving the decomposition into disjoint cosets.

For the $B = NT = TN$, notice that:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} \in TN$$

It can also be factored as:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in NT$$

Showing they are a semidirect products completing the proof.

By constructing, any representative of the coset BwB has a nonzero entry in the $(2, 1)$-position. As $B = TN = NT$ and w normalizes T ($wTw = T$):

$$BwB = NwB = BwN$$

Furthermore, the map:

$$B \times N \to BwN \qquad (b, n) \mapsto bwn$$

is *bijective*.

Next, For any $M \in \mathrm{GL}_2(F)$, define $\mathrm{ch}_M(t) = \det(tI - M) = t^2 - \mathrm{tr}(M)t + \det(M)$, that is $\mathrm{ch}_M(t)$ is the characteristic polynomial of M, i.e. a monic quadratic F-polynomial and $ch_M(0) = \det(M) \neq 0$.

Proposition 1.5.3: Character Polynomial For $\mathrm{GL}_2(F)$

Let F be an arbitrary field, $M \in \mathrm{GL}_2(F)$, and $f(t) = \mathrm{ch}_M(t) = \det(tI - M)$. Then:

1. If $f(t)$ is irreducible, the F-algebra $F[M] \subseteq M_2(F)$ is a field with $\mathrm{GL}_2(F)$-centralizer $F[M]^*$. Given $f(t) = t^2 + at + b$, then M is $\mathrm{GL}_2(F)$-conjugate to:
$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$
Furthermore, an element $N \in \mathrm{GL}_2(F)$ is GL_2-conjugate go M if and only if $\mathrm{ch}_N(t) = f(t)$.

2. If $f(t)$ has distinct roots $a, b \in F^*$, then M is $\mathrm{GL}_2(F)$-conjugate to:
$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

3. If $f(t)$ has a repeated root $a \in F^*$, Then M is $\mathrm{GL}_2(F)$-conjugate to either:
$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$
In the first case, M lies in the center of $\mathrm{GL}_2(F)$, and in the 2nd the $\mathrm{GL}_2(F)$-centralizer of M is $F[M]^*$.

***Proof*:**
We shall do one and leave the rest as an exercise.

1. As $f(t)$ is irreducible, by Cayley-Hamilton $F[M] \cong F[t]/(f(t))$ where the characteristic polynomial $f(t)$ is also the minimal polynomial. For the conjugacy result, as M is irreducible it has no eigenvalues and so $\{v, Mv\}$ forms a basis for any $v \in F^2$. Finding the matrix representation of the linear transformation corresponding to M shall give the desired result

2. exercise

3. exercise

Now let $F = \mathbb{F}_q$ be a field with $q = p^l$ elements for some $l \in \mathbb{N}$, and let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Let us find the irreducible complex representations of G. First off:

Lemma 1.5.4: Order of G

Let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Then:
$$|G| = (q^2 - 1)(q^2 - q)$$
Furthermore, G has $q^2 - 1$ conjugacy classes.

Thus, there are $q^2 - 1$ irreducible representations.

> ***Proof*:**
> Let the first column be any nonzero vector, giving $q^2 - 1$ possibilities. The second column cannot be a scalar multiple of the first. Given a choice v_1 of the first column, the second cannot be kv_1 for all $k \in \mathbb{F}_q$. This gives $q^2 - q$ possibilities. These choices are linearly independent, thus:
>
> $$|G| = (q^2 - 1)(q^2 - q)$$
>
> For the conjugacy classes we repeatedly use proposition 1.5.3. By the first case, as $\mathrm{GL}_2(\mathbb{F}_q)$ has exactly $(q^2 - q)/2$ irreducible monic polynomials of degree 2, there are $(q^2 - q)/2$ conjugacy classes. The second case will give $(q - 1)(q - 2)/2$ cases, and the third $2(q - 1)$ cases. Summing these up gives the desires result.

Thus, there are $q^2 - 1$ irreducible representations. As G is finite, we may take advantage of character theory, hence we shall construct some characters. First, note that $N \cong (\mathbb{F}_q, +)$ via $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

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here

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